Degree and Variable Lower Bounds for Diophantine Definitions

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Abstract

Let K be a number field and let S be a finite set of its finite primes. A single polynomial whose K-rational input projection is an infinite subset of the ring of S-integers must have total degree at least four and at least three total coordinates. The coordinate bound also holds for finite systems of arbitrary degree, even with nonzero conditions and finite alternatives. Systems of quadratic equations require at least five total coordinates; systems of equations of degree at most three require at least four. More precisely, a system in three coordinates with an infinite integral projection must have a common irreducible factor of degree at least four. For a single polynomial, the input and every essential witness coordinate must each have individual degree at least two. The proof combines an elementary argument about prime divisors of polynomial values, Siegel's theorem for integral points on curves, and Kollár's unirationality theorem for cubic hypersurfaces. Projection from a smooth point reduces a surface intersection of two quadrics to a cubic surface, including the necessary reducible and nonreduced cases. Every nonnegative quartic requires at least five coordinates, and a nonnegative polynomial in three coordinates requires degree at least eight. These are necessary coordinate bounds; attainment of the unrestricted three-coordinate bound is not established.

1 Definitions and statements

For a nonzero prime ideal $\mathfrak{p}$ of the ring of integers $\mathcal{O}_K$, write $v_\mathfrak{p}$ for its additive valuation, normalized to have value group $\mathbb{Z}$, and put $v_\mathfrak{p}(0) = +\infty$. For a finite set S of nonzero prime ideals, define

$$\mathcal{O}_{K,S} = \{a \in K : v_\mathfrak{p}(a) \geq 0 \text{ for every } \mathfrak{p} \notin S\}.$$

In particular, $\mathcal{O}_{K,\varnothing} = \mathcal{O}_K$.

Definition 1.1. A single-polynomial presentation over K is an equation

$$P(T, Y_1, \ldots, Y_m) = 0, \qquad P \in K[T, Y_1, \ldots, Y_m].$$

Its *input projection* is the set

$$A_P = \{t \in K : \exists \mathbf{y} \in K^m, P(t, \mathbf{y}) = 0\}.$$

The distinguished input T is counted among the coordinates; thus there are $m+1$ total coordinates and m existential witnesses. Degree means total degree, including T. A witness coordinate is *essential* if the polynomial depends on it.

A finite system uses the same coordinates for all its equations. We also consider finite unions of presentations consisting of equations and nonzero conditions. Geometrically, each such presentation is a Zariski open subset of a common zero set. All coordinates are evaluated in K, and no ordering conditions or universal quantifiers are allowed.

Theorem 1.2. *Let K be a number field and S a finite set of finite primes.*

1. *If a single-polynomial presentation has an infinite input projection contained in $\mathcal{O}_{K,S}$, then it has at least three total coordinates and total degree at least four.*
2. *The same lower bound of three total coordinates holds for finite systems of arbitrary degree, with nonzero conditions and finite alternatives permitted.*
3. *In the single-polynomial case, the individual degree in T and in each essential witness coordinate is at least two.*
4. *A finite system whose equations have degree at most three and whose input projection is infinite and contained in $\mathcal{O}_{K,S}$ requires at least four total coordinates, even with nonzero conditions and finite alternatives.*
5. *A finite system of quadratic equations with an infinite input projection contained in $\mathcal{O}_{K,S}$ requires at least five total coordinates. Nonzero conditions and finite alternatives are permitted.*

For $K = \mathbb{Q}$ and $S = \varnothing$, this applies in particular to every polynomial definition of $\mathbb{Z}$ over $\mathbb{Q}$. The degree restriction in part (1) concerns one polynomial. A system of quadratic equations is a different presentation, and is not excluded by that restriction. In particular, part (5) applies to quartics written as positive sums of squares of quadratic equations. Section 7 extends the five-coordinate bound to every nonnegative quartic over a field with a real embedding. The bound is not asserted here for every quartic.

Two established geometric results enter the proof. Siegel's theorem gives finiteness of integral points on affine curves of positive genus, including S-integral points over a number field [2, Chapter 8]. Kollár's theorem gives unirationality of a noncone cubic hypersurface of dimension at least two over a perfect field as soon as it has a rational point [1, Theorem 2]. All number fields are perfect. The other ingredients are proved below.

2 Rational functions and primes outside a finite set

Lemma 2.1. *Let $g \in \mathcal{O}_K[Z]$ be nonconstant. For every finite set Σ of nonzero prime ideals of $\mathcal{O}_K$, there exist $a \in \mathcal{O}_K$ and $\mathfrak{p} \notin \Sigma$ such that*

$$g(a) \neq 0, \qquad v_{\mathfrak{p}}(g(a)) > 0.$$

Proof. After replacing $g(Z)$ by $g(Z+c)$ for a suitable $c \in \mathcal{O}_K$, we may suppose that $b = g(0) \neq 0$. Choose a nonzero rational integer M belonging to every ideal in Σ; take $M = 1$ if Σ is empty. Write $g(Z) = b + c_1 Z + \cdots + c_d Z^d$, where $d \geq 1$ and $c_d \neq 0$. Then

$$h(N) = \frac{g(bMN)}{b} = 1 + c_1 MN + c_2 bM^2N^2 + \cdots + c_d b^{d-1} M^d N^d \in \mathcal{O}_K[N].$$

For every rational integer n and every $\mathfrak{q} \in \Sigma$, we have $h(n) \equiv 1 \pmod{\mathfrak{q}}$. The norm polynomial

$$\prod_{\sigma : K \hookrightarrow \mathbb{C}} h^{\sigma}(N)$$

lies in $\mathbb{Z}[N]$ and has degree $d[K:\mathbb{Q}]$: its leading coefficient is the nonzero norm of $c_d b^{d-1} M^d$. Its absolute value is therefore greater than one for all sufficiently large positive integers n. Such an $h(n)$ is nonzero and is not a unit of $\mathcal{O}_K$. A prime ideal dividing it lies outside Σ and also divides the nonzero value $g(bMn) = bh(n)$. Undoing the initial translation gives the required a. □

Lemma 2.2. *Let $r \in K(Z)$ be nonconstant, and let $E \subset K$ be finite. There is an $a \in K \setminus E$ at which r is defined and $r(a) \notin \mathcal{O}_{K,S}$.*

Proof. First suppose that r is a polynomial. Choose a prime ideal $\mathfrak{p} \notin S$ at which all its coefficients are integral and its leading coefficient is a unit. Only finitely many primes are excluded by these conditions. Choose $0 \neq \pi \in \mathfrak{p}$. For $a = \pi^{-n}$ and sufficiently large n, the leading term of $r(a)$ has strictly smaller $\mathfrak{p}$-adic valuation than all other terms. Its valuation is negative, so $v_{\mathfrak{p}}(r(a)) < 0$. The numbers π^{-n} are distinct, and hence all but finitely many avoid E.

Otherwise write $r = f/g$ with relatively prime polynomials $f, g \in \mathcal{O}_K[Z]$ and with g nonconstant. The polynomial Euclidean algorithm over K and clearing denominators give $A, B \in \mathcal{O}_K[Z]$ and $0 \neq c \in \mathcal{O}_K$ satisfying

$$Af + Bg = c.$$

Enlarge S to a finite set Σ containing the prime ideals dividing c or any nonzero value $g(e)$ with $e \in E \cap \mathcal{O}_K$. By Lemma 2.1, there are $a \in \mathcal{O}_K$ and $\mathfrak{p} \notin \Sigma$ with $g(a) \neq 0$ and $v_{\mathfrak{p}}(g(a)) > 0$. The displayed identity implies $v_{\mathfrak{p}}(f(a)) = 0$. Thus $r(a)$ has negative valuation at $\mathfrak{p} \notin S$. Moreover $a \notin E$: if $a = e$ were in $E \cap \mathcal{O}_K$, the prime divisor $\mathfrak{p}$ of its nonzero value $g(e)$ would have been included in Σ. □

An integral K-variety is called *K-unirational* if it is dominated by an affine space through a rational map defined over K.

Proposition 2.3. *Let X be a K-unirational integral variety, let $U \subseteq X$ be nonempty and open, and let $f : U \to \mathbb{A}^1_K$ be regular. If*

$$f(U(K)) \subseteq \mathcal{O}_{K,S},$$

then f is constant.

Proof. Pull back f through a dominant rational parametrization. If f is nonconstant, the result is a nonconstant rational function on affine space. Shrink its permitted domain to a nonempty principal open subset. In characteristic zero, one of its partial derivatives is nonzero. Since K is infinite, we can fix the other coordinates in K so that this derivative remains nonzero and none of the polynomials defining the permitted open set vanishes identically on the resulting line. The restriction is a nonconstant rational function $r \in K(Z)$, and the parametrization is permitted at every K-point of the line except a finite set. Lemma 2.2 then supplies a value outside $\mathcal{O}_{K,S}$, a contradiction. □

3 Integral-valued functions on curves

Lemma 3.1. *Let C be a smooth projective geometrically integral curve over K of positive genus, and let $f \in K(C)$ be nonconstant. There are only finitely many $P \in C(K)$ at which f is defined and $f(P) \in \mathcal{O}_{K,S}$.*

Proof. The function f defines a finite morphism $C \to \mathbb{P}^1_K$. Removing its poles gives an affine curve $D = f^{-1}(\mathbb{A}^1_K)$ finite over $\mathbb{A}^1_K$. Choose generators $z_1, \ldots, z_r$ of its coordinate ring over $K[f]$. Each z_i satisfies a monic equation over $K[f]$.

There is a finite set S' of finite primes, containing S, such that all coefficients in these finitely many equations belong to $\mathcal{O}_{K,S'}[f]$. If $P \in D(K)$ and $f(P) \in \mathcal{O}_{K,S}$, then each $z_i(P) \in K$ is integral over $\mathcal{O}_{K,S'}$. That ring is integrally closed, so all $z_i(P)$ belong to it. Consequently these points are S'-integral in a fixed affine embedding of D. Siegel's theorem gives their finiteness [2, Chapter 8]. Its positive-genus hypothesis holds for D, and at least one pole was removed because f is nonconstant. □

Proposition 3.2. *Let X be an algebraic set over K of dimension at most one, let $U \subseteq X$ be open, and let $f : U \to \mathbb{A}^1_K$ be regular. If $f(U(K)) \subseteq \mathcal{O}_{K,S}$, then $f(U(K))$ is finite.*

Proof. Pass to the finitely many irreducible components, omitting components disjoint from U. Zero-dimensional components contribute finitely many points. If an irreducible curve is not geometrically integral, every K-point lies in the intersection of distinct conjugate geometric components. These intersections form a finite set, so this case also contributes only finitely many points.

For a geometrically integral curve, take its smooth projective normalization C. Away from finitely many points the normalization is an isomorphism, and f pulls back to a rational function on C. A constant function has finite image. If the genus is positive, Lemma 3.1 applies. If the genus is zero and $C(K)$ is empty, only the finitely many points omitted in passing to the normalization can contribute. If the genus is zero and $C(K)$ is nonempty, then $C \simeq \mathbb{P}^1_K$; Proposition 2.3 excludes a nonconstant function with the asserted integral-value property on the permitted open subset. This proves the result. □

4 Hypersurfaces of degree at most three

Theorem 4.1. *Let $X \subseteq \mathbb{P}^N_K$ be the zero set of a nonzero homogeneous polynomial of degree at most three. For every open $U \subseteq X$ and regular function $f : U \to \mathbb{A}^1_K$, the containment $f(U(K)) \subseteq \mathcal{O}_{K,S}$ implies that $f(U(K))$ is finite. The same conclusion holds for $X = \mathbb{P}^N_K$.*

Proof. Projective space is unirational, so that case follows from Proposition 2.3. Factor the equation of a hypersurface over K and treat each irreducible factor separately. Multiplicities do not change the rational point set.

Geometrically reducible components. An irreducible polynomial of degree two or three over a perfect field, if not geometrically irreducible, splits geometrically into conjugate linear factors. Indeed the geometric factors have a common degree and form one Galois orbit; that degree divides the prime number two or three. Every K-point belongs to all the conjugate hyperplanes. Their common intersection is a K-linear subspace, and Proposition 2.3 applies to the restriction of f. Linear components are treated by the same proposition.

Geometrically integral quadrics. A quadric with a smooth K-point is rational by projection from that point. If it has no smooth K-point, all its K-points lie in its singular locus, which is a linear subspace. Both cases give the required finite image.

Cubics that are not cones. Curves and zero-dimensional cases are covered by Proposition 3.2. A geometrically integral cubic of dimension at least two that is not a cone is K-unirational if it has a K-point, by Kollár's theorem [1, Theorem 2]. This theorem applies to singular cubics as well as smooth ones. If there is no K-point the image is empty; otherwise use Proposition 2.3.

The structure of a cubic cone. Write its cubic equation as F on a K-vector space V and put

$$W = \{w \in V : \partial_w F = 0 \text{ as a polynomial}\}.$$

This is the kernel of a K-linear map. In characteristic zero, F is constant along the directions in W, so it descends to a cubic form on V/W. The projective space $\mathbb{P}(W)$ is the maximal vertex. Choose a K-linear complement to W. It gives a geometrically integral cubic base B which is not a cone. Indeed, a vertex direction of the base would lift to an additional direction in the kernel defining W. That kernel commutes with extension of the ground field, so the same assertion holds geometrically. Its dimension is at least one, since a zero-dimensional cubic hypersurface is not geometrically integral.

Outside $\mathbb{P}(W)$, projection from the vertex gives a morphism to B. Each fiber over a K-point of B is an affine space of dimension $\dim W$. On an open subset of B this bundle is trivial, so the cone is birational to $B \times \mathbb{A}^{\dim W}$. These statements follow directly by writing a point of V as the sum of a vector in W and a vector in the chosen complement.

If $B(K)$ is empty, all rational points of the cone lie in the vertex. If $B(K)$ is finite, apply Proposition 2.3 on the permitted open part of each affine-space fiber, and also on the vertex.

This gives a finite union of finite images. If $\dim B \geq 2$ and $B(K)$ is nonempty, Kollár's theorem makes B, and then the cone, unirational.

A cone over a curve with infinitely many rational points. The remaining case has $\dim B = 1$ and $B(K)$ infinite. Let C be the smooth projective normalization of B. Discarding finitely many base points, write the function f on an open subset of $C \times \mathbb{A}^{\dim W}$. If this rational function depends nontrivially on the affine-space coordinates, some partial derivative in those coordinates is nonzero. Expressing it as a quotient of polynomials with coefficients in $K(C)$ shows that, outside finitely many base points, the specialized function is nonconstant and its permitted fiber is nonempty. Choose a K-point of C outside those exceptions. Its fiber is a nonempty open subset of affine space, contradicting Proposition 2.3. Therefore f belongs to $K(C)$.

For every K-point of C outside a finite set, the permitted fiber is a nonempty open subset of affine space and has a K-point. Thus the descended function is defined and S-integral at all these base points. Proposition 3.2 implies that its image there is finite. Since a nonconstant rational function on a curve has finite fibers and $C(K)$ is infinite, the function must be constant. It is then constant as an element of the cone's function field, and hence wherever it is regular on that irreducible cone. This includes the points omitted from the chosen birational chart. □

5 Consequences for polynomial presentations

Proof of Theorem 1.2(1) and (2). First consider at most two total coordinates. For one system of equations, its zero set is either all of the ambient affine space or has dimension at most one. In the first case, the permitted nonempty open subset is unirational, and its input function cannot have an infinite S-integral image by Proposition 2.3. In the second case, Proposition 3.2 gives a finite input image. Empty permitted sets contribute nothing. Taking finitely many alternatives preserves finiteness. This proves the coordinate bound for systems and for single equations.

For a single nonconstant polynomial of total degree at most three, homogenize it and regard the distinguished input as a regular function on the usual affine chart of its projective zero set. Theorem 4.1 gives a finite input image, independently of the number of coordinates. A nonzero constant polynomial has empty projection. The zero polynomial has projection K, which is not contained in $\mathcal{O}_{K,S}$ because S is finite. These observations prove the degree bound. □

Proposition 5.1. *Suppose that $P \in K[T, Y_1, \ldots, Y_m]$ has an infinite input projection contained in $\mathcal{O}_{K,S}$. Then*

$$\deg_T P \geq 2, \qquad \deg_{Y_j} P \geq 2 \quad \textit{for every essential witness } Y_j.$$

Proof. Suppose first that P is linear in an essential witness Y_j. Write

$$P = A(T, \mathbf{Y}')Y_j + B(T, \mathbf{Y}'), \qquad A \neq 0.$$

Choose constants $\mathbf{c} \in K^{m-1}$ so that $A(T, \mathbf{c})$ is a nonzero polynomial. Such a choice exists because K is infinite. For every t outside its finite root set, setting $Y_j = -B(t, \mathbf{c})/A(t, \mathbf{c})$ gives a rational witness. The projection is therefore cofinite in K. It cannot be contained in $\mathcal{O}_{K,S}$, whose complement is infinite: for a prime $\mathfrak{p} \notin S$ and $0 \neq \pi \in \mathfrak{p}$, all the distinct elements π^{-n} lie outside $\mathcal{O}_{K,S}$.

Suppose next that $\deg_T P \leq 1$, and write $P = A(\mathbf{Y})T + B(\mathbf{Y})$. If $A = 0$, the input projection is either empty or all of K. Otherwise on the nonempty open set $A \neq 0$ the rational function $-B/A$ takes only values in $\mathcal{O}_{K,S}$. Proposition 2.3 makes it constant. At any rational point with $A = B = 0$, every input would be admitted, which is impossible. Thus the projection has at most one element, again a contradiction. This proves Theorem 1.2(3). □

Corollary 5.2. *No finite system of linear equations, even with nonzero conditions and finite alternatives, has an infinite input projection contained in $\mathcal{O}_{K,S}$.*

Proof. Each nonempty common zero set of linear equations is an affine space over K. Apply Proposition 2.3 to its permitted open subset and take the finite union of the resulting finite images. □

Proposition 5.3. *Let $F_1, \dots, F_s \in K[T, Y_1, Y_2]$, and let U be a Zariski open subset of their common zero set. Suppose that $T(U(K))$ is infinite and contained in $\mathcal{O}_{K,S}$. Then some irreducible polynomial of degree at least four divides every F_i. In particular every nonzero F_i has degree at least four. The nonzero conditions defining U may have arbitrary degrees.*

Proof. The case in which all F_i vanish identically is impossible: then U is a nonempty open subset of affine three-space, and Proposition 2.3 would make its input function constant. Discard any identically zero equations and put $g = \gcd(F_1, \dots, F_s)$, computed in the unique factorization domain $K[T, Y_1, Y_2]$. Write $F_i = gh_i$. Then

$$V(F_1, \dots, F_s) = V(g) \cup V(h_1, \dots, h_s).$$

The second zero set has dimension at most one. Indeed, an irreducible component of dimension two would correspond to a height-one prime of the polynomial ring. Such a prime is generated by one irreducible polynomial, which would divide all the h_i, contrary to $\gcd(h_1, \dots, h_s) = 1$. A component of dimension three would make all the h_i zero, which is also impossible.

By Proposition 3.2, the permitted open subset of $V(h_1, \dots, h_s)$ has finite input image. Factor g into irreducibles. There are finitely many resulting hypersurfaces, so at least one factor g_0 satisfies

$$T\big((U \cap V(g_0))(K)\big) \quad \text{is infinite.}$$

Every such input lies in $\mathcal{O}_{K,S}$. If $\deg g_0 \leq 3$, homogenizing g_0 and applying Theorem 4.1 to this permitted affine open subset gives a finite image, a contradiction. Therefore $\deg g_0 \geq 4$. Since g_0 divides g, it divides every defining equation. □

Corollary 5.4. *A finite system of equations of total degree at most three with an infinite input projection contained in $\mathcal{O}_{K,S}$ uses at least four total coordinates. The same conclusion holds with arbitrary nonzero conditions and finitely many alternatives. For quadratic systems, Section 6 gives the stronger bound of five total coordinates.*

Proof. If there are fewer than three coordinates, adjoining unused coordinates gives a presentation in three coordinates with the same input projection and the same equation degrees. Proposition 5.3 excludes an infinite image for each alternative. A finite union of finite images is finite. This proves Theorem 1.2(4). □

6 Quadratic systems in four coordinates

The extra coordinate obstruction for quadratic systems follows from an explicit projection. The multiplicity issue matters: the common zero set of two quadrics need not be reduced.

Lemma 6.1. *Let $Y \subseteq \mathbb{P}^4_K$ be a geometrically integral projective surface of degree at most three. For every open $V \subseteq Y$ and regular function $f : V \to \mathbb{A}^1_K$, if $f(V(K)) \subseteq \mathcal{O}_{K,S}$, then $f(V(K))$ is finite.*

Proof. The elementary degree inequality

$$\deg Y \geq 1 + \operatorname{codim}(Y, \langle Y \rangle)$$

for a projective integral variety is recalled in [3, Introduction, inequality $(*)$]. The linear span $\langle Y \rangle$ is defined over K. If the span has dimension at most three, Y is a plane or a hypersurface of degree at most three in that span, so Theorem 4.1 applies.

The remaining case has $\deg Y = 3$ and $\langle Y \rangle = \mathbb{P}^4$. If $V(K)$ is not dense, its closure has dimension at most one, and Proposition 3.2 applies. Otherwise choose a smooth point $P \in V(K)$. Projection from P has a surface as its image: if the image had dimension one, Y would be a cone with vertex P; every generating line would then lie in the projective tangent plane at the smooth point P, contradicting the span. Write $Y' \subseteq \mathbb{P}^3$ for the image surface, and e for the generic degree of the projection. The image is nondegenerate, because a hyperplane containing it would pull back to a hyperplane containing Y. Intersecting with a general line in $\mathbb{P}^3$, whose inverse image is a plane through P, gives

$$3 = 1 + e \deg Y'.$$

Here the contribution at P is one because P is smooth and the plane is general. This is the point-projection degree formula [5, Theorem 5.11], with multiplicity one. A nondegenerate surface in $\mathbb{P}^3$ has degree at least two; thus $e = 1$ and Y' is a quadric. On isomorphic nonempty open subsets the rational function f therefore corresponds to a regular function on an open subset of that quadric. Theorem 4.1 gives finite image there. The omitted subset of Y has dimension at most one, and Proposition 3.2 handles its image. □

Proposition 6.2. *Let $Q_1, Q_2 \in K[X_0, \dots, X_4]$ be relatively prime homogeneous quadratic forms. Let Y be an irreducible surface component of their projective common zero set, endowed with its reduced structure. If $V \subseteq Y$ is open and $f : V \to \mathbb{A}^1_K$ is regular with $f(V(K)) \subseteq \mathcal{O}_{K,S}$, then $f(V(K))$ is finite.*

Proof. If Y is not geometrically integral, all its K-points lie in the intersections of distinct conjugate geometric components. Their union descends to a proper closed subset of Y of dimension at most one. Proposition 3.2 proves the assertion in this case. Hence assume that Y is geometrically integral.

If $V(K)$ is not Zariski dense in Y, its Zariski closure is again of dimension at most one, and the same proposition applies. Suppose therefore that $V(K)$ is dense. Bézout's theorem for the complete intersection $Z = V(Q_1, Q_2) \subset \mathbb{P}^4_K$ gives

$$\sum_i m_i \deg Y_i = 4,$$

where Y_i are the geometric surface components and m_i their multiplicities at their generic points. If the multiplicity of Y is at least two, then $\deg Y \leq 2$, and Lemma 6.1 applies.

It remains to treat multiplicity one. The complete intersection Z is then reduced at the generic point of Y. Over the perfect field K, there is a nonempty open subset of Y on which Z is smooth of dimension two and no other component occurs. Density supplies a point $P \in V(K)$ in that open subset. After a K-linear projective change of coordinates, put $P = (1 : 0 : 0 : 0 : 0)$. Because $Q_i(P) = 0$, write

$$Q_1 = X_0 L_1(\mathbf{X}) + q_1(\mathbf{X}), \qquad Q_2 = X_0 L_2(\mathbf{X}) + q_2(\mathbf{X}), \qquad \mathbf{X} = (X_1, \dots, X_4),$$

where the L_i are linear and the q_i quadratic. Smoothness of Z at P says that L_1, L_2 are linearly independent.

If both L_i vanish identically on Y, then Y is the plane $V(L_1, L_2)$, and Lemma 6.1 applies. Otherwise, after interchanging the two equations, assume that L_1 does not vanish identically on Y. Projection from P then has the rational inverse

$$[X_1 : \cdots : X_4] \longmapsto [-q_1(\mathbf{X}) : L_1(\mathbf{X})X_1 : \cdots : L_1(\mathbf{X})X_4]$$

on the open set $L_1 \neq 0$. The image lies in the cubic

$$H = L_1 q_2 - L_2 q_1 = 0 \quad \text{in } \mathbb{P}^3_K.$$

This polynomial is nonzero. Indeed, if $H = 0$, relative primality of the independent linear forms would give $q_1 = L_1 M$ and $q_2 = L_2 M$ for one linear form M. Both quadrics would then be divisible by $X_0 + M$, contrary to their relative primality.

The image closure of Y is thus a surface component Y' of the cubic $V(H)$. It is defined by an irreducible factor of H, of degree at most three. The displayed inverse gives an isomorphism between nonempty open subsets of Y and Y'. Shrink the first open subset so that it lies in V. On the corresponding open subset of Y', the function f is a *rational* function, regular after this shrinking, and all its K-rational values lie in $\mathcal{O}_{K,S}$. Theorem 4.1 makes its image finite. The omitted closed subset of the surface Y has dimension at most one; Proposition 3.2 makes the remaining image finite. □

Theorem 6.3. *Let $F_1, \dots, F_s \in K[T, Y_1, Y_2, Y_3]$ have total degree at most two, and let U be open in their common zero set. If $T(U(K)) \subseteq \mathcal{O}_{K,S}$, then $T(U(K))$ is finite. Consequently every finite quadratic system with infinite S-integral input projection needs at least five total coordinates. The assertion is unchanged by nonzero conditions or finitely many alternatives.*

Proof. Identically zero equations may be discarded. If all equations are zero, use Proposition 2.3; a nonzero constant equation gives the empty set. A nonconstant linear equation allows a K-affine parametrization of its hyperplane by three coordinates. If the input is constant on that hyperplane, the claim is immediate. Otherwise choose the parametrization with the input itself as one of those coordinates. The remaining equations still have degree at most two, so Corollary 5.4 applies. We may therefore assume that all remaining equations have degree exactly two.

If $g = \gcd(F_1, \dots, F_s)$ is nonconstant, write $F_i = g h_i$. Their common zero set is

$$V(g) \ \cup \ V(h_1, \dots, h_s).$$

The first part has finite permitted input image by Theorem 4.1. The h_i have degree at most one, so the second part has finite input image by Corollary 5.2.

Suppose now that the gcd is one. Choose one nonzero equation F. A K-linear combination G of the equations can be chosen relatively prime to F: for each of the finitely many irreducible factors of F, divisibility of the combination by that factor is a proper linear condition on its coefficients. A finite union of proper linear subspaces cannot cover the coefficient space over the infinite field K. The case $\deg G \leq 1$ is already settled, so take $\deg G = 2$.

The homogenizations Q_1, Q_2 of F, G in $\mathbb{P}^4_K$ remain relatively prime. A common factor other than the homogenizing coordinate would dehomogenize to a common factor of F, G; neither homogenization is divisible by that coordinate. Their intersection is therefore a complete intersection of pure dimension two. Every surface component of the projective closure of $V(F_1, \dots, F_s)$ is a component of this intersection. Proposition 6.2 makes its permitted input image finite. Components of dimension at most one are covered by Proposition 3.2. There are only finitely many components, proving the assertion.

Adjoining unused coordinates treats systems in fewer than four coordinates, and a finite union of finite images is finite. This proves Theorem 1.2(5). □

Corollary 6.4. *Let K have a real embedding σ, let $c_1, \dots, c_s \in K$ satisfy $\sigma(c_i) > 0$, and let $F_i \in K[T, \mathbf{Y}]$ have degree at most two. If*

$$P = \sum_{i=1}^{s} c_i F_i^2$$

has infinite input projection contained in $\mathcal{O}_{K,S}$, then P uses at least five total coordinates.

Proof. For a K-rational tuple, $P = 0$ holds if and only if every $F_i = 0$, as follows by applying σ. Apply Theorem 6.3. □

7 Three-coordinate presentations and nonnegative polynomials

Proposition 7.1. *Suppose that $P \in K[T, Y_1, Y_2]$ has infinite input projection contained in $\mathcal{O}_{K,S}$. Then P has an absolutely irreducible factor G of degree at least four such that $V(G)(K)$ is Zariski dense and its input projection is infinite. Every proper closed subset of $V(G)$ has finite input projection on its K-points. In particular, if $\deg P = 4$, then P is absolutely irreducible up to a nonzero constant factor and its smooth K-points have infinite input projection.*

Proof. Factor P over K. Some irreducible factor G has infinite input projection, since a finite union of finite sets is finite. If G is not absolutely irreducible, its rational points lie in the intersections of conjugate geometric components, a closed subset of dimension at most one. Proposition 3.2 would give a finite image. Hence G is absolutely irreducible. The same proposition shows that no proper closed subset of its surface can contain all rational points contributing infinitely many inputs. Thus the rational points are dense. The degree bound follows from Theorem 4.1. If $\deg P = 4$, the factor G already has the full degree and occurs once. Its singular locus is a proper closed subset of dimension at most one, hence contributes only finitely many inputs. □

Theorem 7.2. *Let K have a real embedding σ. Suppose that $P \in K[T, Y_1, Y_2]$ satisfies*

$$P^\sigma(t, y_1, y_2) \geq 0 \quad \text{for all } (t, y_1, y_2) \in \mathbb{R}^3.$$

If its K-rational input projection is infinite and contained in $\mathcal{O}_{K,S}$, then P has a repeated absolutely irreducible factor of degree at least four. Consequently $\deg P \geq 8$.

Proof. Choose the factor G of Proposition 7.1. Every real zero of a nonnegative differentiable function is a local minimum, so all first partial derivatives of P^σ vanish at every point of $V(G)(K)$ after applying σ. The embedding is injective. Therefore all first partial derivatives of P vanish on the Zariski-dense set $V(G)(K)$, and G divides each of them.

Write $P = G^e H$ with $G \nmid H$. If $e = 1$, reduction of $\partial P/\partial X_j$ modulo G gives

$$H \frac{\partial G}{\partial X_j} = 0 \quad \text{in } K[T, Y_1, Y_2]/(G)$$

for each coordinate X_j. This quotient is an integral domain and H is nonzero in it, so G divides every partial derivative of itself. Their degrees are smaller than $\deg G$, forcing every derivative to be zero. In characteristic zero this would make G constant, a contradiction. Thus $e \geq 2$, and $\deg P \geq 2 \deg G \geq 8$. □

Corollary 7.3. *A three-coordinate quartic defining an infinite subset of $\mathcal{O}_{K,S}$ must take both positive and negative values at every real embedding of K. In particular, a quartic over $\mathbb{Q}$ whose rational input projection is $\mathbb{Z}$ cannot be nonnegative on $\mathbb{R}^3$.*

Proof. Otherwise the real polynomial has a constant sign, and either P or $-P$ contradicts Theorem 7.2. □

Lemma 7.4. *Let $X \subset \mathbb{P}^4_K$ be a geometrically integral quartic hypersurface. Every geometric surface component of its singular locus has degree at most three.*

Proof. Work over an algebraic closure of K, and let Y be such a component. By Bertini's irreducibility theorem, applied twice, a general plane $\Lambda \simeq \mathbb{P}^2$ cuts X in an integral plane quartic C; see [6, Lemma 37.32.3]. The same general plane meets the reduced surface Y transversely at $\deg Y$ distinct points. At each such point all partial derivatives of the quartic equation of X vanish, so the restricted plane equation is singular there.

The arithmetic genus of a plane quartic is $(4-1)(4-2)/2 = 3$. If $\widetilde{C}$ is its normalization, the normalization exact sequence gives

$$3 = g(\widetilde{C}) + \sum_{P \in \mathrm{Sing}(C)} \delta_P, \qquad \delta_P = \dim_{\overline{K}}(\widetilde{\mathcal{O}}_{C,P}/\mathcal{O}_{C,P}).$$

Every singular point of an integral plane curve has $\delta_P \geq 1$, and the geometric genus is nonnegative. Thus C has at most three singular points, and $\deg Y \leq 3$. □

Theorem 7.5. *Let K have a real embedding σ and let $P \in K[T, Y_1, Y_2, Y_3]$ have degree at most four. Suppose that P^σ is nonnegative on $\mathbb{R}^4$. If the rational input projection of P is contained in $\mathcal{O}_{K,S}$, it is finite. Consequently a nonnegative quartic with infinite S-integral input projection needs at least five total coordinates.*

Proof. The zero polynomial is excluded by the assumed projection containment, while a nonzero constant has empty projection. If all irreducible factors of P over K have degree at most three, each factor has its rational zero set contained in that of P, so Theorem 4.1 gives finite input image for each. It therefore suffices to consider an irreducible quartic.

Suppose first that the quartic is not absolutely irreducible. In characteristic zero its geometric factors are distinct, have equal degrees, and form one Galois orbit. Thus it is a nonzero constant times either the norm of a quadratic polynomial over a quadratic extension or the norm of a linear polynomial over a degree-four extension. To see the precise field of definition, normalize one nonzero coefficient of a geometric factor to one; its stabilizer then fixes all its coefficients, and its orbit has respectively two or four elements. On a K-rational tuple the norm vanishes exactly when that polynomial vanishes. Expanding in a K-basis of the extension gives respectively two quadratic equations or four linear equations over K. Their input image is finite by Theorem 6.3 or Corollary 5.2.

Finally suppose that P is absolutely irreducible, and let $X \subset \mathbb{P}^4_K$ be the quartic obtained by homogenization. Every rational affine zero of P is a real minimum after applying σ. All its first derivatives vanish. Euler's identity for the homogenized polynomial shows that its derivative in the homogenizing coordinate also vanishes. Therefore all rational affine zeros belong to $\mathrm{Sing}(X)$.

This singular locus has dimension at most two. Components of dimension at most one contribute finite input image by Proposition 3.2. If a surface component is not geometrically integral, its rational points lie in a closed subset of dimension at most one, with the same conclusion. Every geometrically integral surface component has degree at most three by Lemma 7.4; its input image on the affine chart is finite by Lemma 6.1. Taking the finite union over components proves the theorem. □

8 Scope of the coordinate bounds

Corollary 8.1. *Let $A \subseteq \mathcal{O}_{K,S}$ be infinite. If A has a single-polynomial presentation of degree four over K, then four is the least possible degree of any such presentation, regardless of how many witness coordinates are allowed. If A has a presentation by a system of quadratic equations, then two is the least possible maximum degree of the equations in a system.*

Proof. The first assertion follows from Theorem 1.2; the second follows from Corollary 5.2. □

The bounds for systems can be displayed as follows; the degree in the table is the largest degree among the equations.

maximum equation degree	necessary total coordinates
1	no finite number suffices
2	at least 5
3	at least 4
4 or greater	at least 3

The nonzero conditions may have arbitrary degrees. For a single equation, degrees at most three are impossible regardless of the number of coordinates. All nonnegative quartics, including positive sums of squares of quadratics, need at least five coordinates, while arbitrary quartics have only the unconditional three-coordinate lower bound proved here. Nonnegative polynomials in three coordinates need degree at least eight.

Let ν be the least total number of coordinates in a single-polynomial definition of $\mathbb{Z}$ over $\mathbb{Q}$, if such a definition exists, and put $\nu = +\infty$ otherwise. The unconditional conclusion is

$$\nu \geq 3.$$

An upper bound obtained from a construction is subject to that construction's stated arithmetic hypotheses. The proofs above do not establish $\nu = 3$, do not exclude $\nu = 3$, and do not provide an attained minimum for quartics or quadratic systems. In particular, the sharp lower bound on *degree* supplied by a valid quartic construction does not make its number of coordinates optimal.

The distinction between the number of quantified coordinates and the dimension of the witness fibers is also essential. For terminology and general results concerning existential rank and essential fiber dimension, see [4]. In the convention of the present paper, the total coordinate count is one more than the number of existentially quantified coordinates.

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